

LOCATION AND TRANSPORTATION CREDIT

Green Vehicles

This credit applies to:

New Construction (1 point)
Core and Shell (1 point)
Data Centers (1 point)
Hospitality (1 point)
Retail (1 point)
Healthcare (1 point)
Schools (1 point)
Warehouses and Distribution Centers (1 point)

INTENT

To reduce pollution by promoting alternatives to conventionally fueled automobiles.

REQUIREMENTS

NEW CONSTRUCTION, CORE AND SHELL, DATA CENTERS, HOSPITALITY, RETAIL, HEALTHCARE

Designate 5% of all parking spaces used by the project as preferred parking for green vehicles. Clearly identify and enforce for sole use by green vehicles. Distribute preferred parking spaces proportionally among various parking sections (e.g. between short-term and long-term spaces).

Green vehicles must achieve a minimum green score of 45 on the American Council for an Energy Efficient Economy (ACEEE) annual vehicle rating guide (or local equivalent for projects outside the U.S.).

A discounted parking rate of at least 20% for green vehicles is an acceptable substitute for preferred parking spaces. The discounted rate must be publicly posted at the entrance of the parking area and permanently available to every qualifying vehicle.

In addition to preferred parking for green vehicles, meet one of the following two options for alternative-fuel fueling stations:

OPTION 1. ELECTRIC VEHICLE CHARGING

Install electrical vehicle supply equipment (EVSE) in 2% of all parking spaces used by the project. Clearly identify and reserve these spaces for the sole use by plug-in electric vehicles. EVSE parking spaces must be provided in addition to preferred parking spaces for green vehicles.

The EVSE must:

- Provide a Level 2 charging capacity (208 – 240 volts) or greater.
- Comply with the relevant regional or local standard for electrical connectors, such as SAE Surface Vehicle Recommended Practice J1772, SAE Electric Vehicle Conductive Charge Coupler or IEC 62196 of the International Electrotechnical Commission for projects outside the U.S.
- Be networked or internet addressable and be capable of participating in a demand-response program or time-of-use pricing to encourage off-peak charging.

OR

OPTION 2. LIQUID, GAS, OR BATTERY FACILITIES

Install liquid or gas alternative fuel fueling facilities or a battery switching station capable of refueling a number of vehicles per day equal to at least 2% of all parking spaces.

SCHOOLS

OPTION 1. GREEN PASSENGER VEHICLES

Designate 5% of all parking spaces used by the project as preferred parking for green vehicles. Clearly identify and enforce for sole use by green vehicles. Distribute preferred parking spaces proportionally among various parking sections (e.g. between short-term and long-term spaces).

Green vehicles must achieve a minimum green score of 45 on the American Council for an Energy Efficient Economy (ACEEE) annual vehicle rating guide (or local equivalent for projects outside the U.S.)

A discounted parking rate of at least 20% for green vehicles is an acceptable substitute for preferred parking spaces. The discounted rate must be publicly posted at the entrance of the parking area and permanently available to every qualifying vehicle.

In addition to preferred parking for green vehicles, meet one of the following two options for alternative-fuel fueling stations:

Path 1. Electric Vehicle Charging

Install electrical vehicle supply equipment (EVSE) in 2% of all parking spaces used by the project. Clearly identify and reserve these spaces for the sole use by plug-in electric vehicles. EVSE parking spaces must be provided in addition to preferred parking spaces for green vehicles.

The EVSE must:

- Provide a Level 2 charging capacity (208 – 240 volts) or greater.
- Comply with the relevant regional or local standard for electrical connectors, such as SAE Surface Vehicle Recommended Practice J1772, SAE Electric Vehicle Conductive Charge Coupler or IEC 62196 of the International Electrotechnical Commission for projects outside the U.S.
- Be networked or internet addressable and be capable of participating in a demand-response program or time-of-use pricing to encourage off-peak charging.

OR

Path 2. Liquid, gas, or battery facilities

Install liquid or gas alternative fuel fueling facilities or a battery switching station capable of refueling a number of vehicles per day equal to at least 2% of all parking spaces.

OR

OPTION 2. GREEN BUSES AND SCHOOL-OWNED VEHICLES

Develop and implement a plan for every bus serving the school to meet the following emissions standards within seven years of the building certificate of occupancy:

- nitrogen oxide (NO_x) emissions of 0.50 grams or less per brake horsepower-hour; and
- particulate matter emissions of 0.01 grams or less per brake horsepower-hour.

Emission standards must be met for each bus and not by an average of the entire fleet serving the school.

Develop and implement a plan for 100% of all other (non-bus) vehicles owned or leased to serve the school to be green vehicles. Green vehicles must achieve a minimum green score of 45 on the American Council for an Energy Efficient Economy (ACEEE) annual vehicle rating guide (or local equivalent for projects outside the U.S).

WAREHOUSES AND DISTRIBUTION CENTERS

OPTION 1. ALTERNATIVE-FUEL VEHICLES (1 POINT)

Provide an on-site fleet with at least one yard tractor that is powered by electricity, propane, or natural gas. Provide on-site charging or refueling stations for the vehicles. Liquid or gas refueling stations must be separately ventilated or located outdoors.

OR

OPTION 2. REDUCED TRUCK IDLING (1 POINT)

Provide an electrical connection for at least 50% of all dock door locations to limit truck idling at the dock.

BEHIND THE INTENT

In 2010, transportation accounted for 27% of the total U.S. greenhouse gas emissions from combustion of petroleum-based fuels. More than half of those emissions came from passenger vehicles and light-duty trucks.[1] Globally, greenhouse gas emissions from transportation increased 17.5% from 1990 to 2010.[2]

Beyond climate change effects, conventional-fuel vehicles pose public health risks. Diesel exhaust from idling buses releases fine particulates, which can be especially harmful for children.[3] Idling delivery trucks at warehouse loading docks and diesel-powered yard tractors used to move cargo containers[4] emit nitrogen oxides (NO_x), sulfur oxides (SO_x), and particulate matter (PM).

This credit addresses vehicle emissions by rewarding local and regional infrastructure that incentivizes the purchase of green vehicles. The provision of preferred parking spaces closest to a building's entrance rewards building users who drive green vehicles. To qualify for these spaces, green vehicles must meet third-party standards that comprehensively measure both fuel efficiency and emissions. Projects must also provide infrastructure (e.g., electric charging stations) for alternative-fuel vehicles. Electric vehicle supply equipment must have effective charging speeds, align with local or regional standards, and take advantage of future improvements to the utility grid.

School projects may address employees' and visitors' vehicle use, as in other rating systems, or have the option of meeting emissions and green vehicle standards for their bus and non-bus vehicle fleets. Warehouses and distribution center projects may purchase alternative-fuel yard tractors to move trailers around the facility or provide electrical connections at loading dock doors, a strategy that allows truck drivers to operate in-cab comfort settings and appliances by plugging into grid power rather than by idling their engines unnecessarily.

STEP-BY-STEP GUIDANCE

NEW CONSTRUCTION, CORE AND SHELL, DATA CENTERS, HOSPITALITY, RETAIL, HEALTHCARE

STEP 1. DETERMINE TOTAL VEHICLE PARKING CAPACITY

Total vehicle parking spaces must include all new and existing off-street parking spaces leased or owned by the project, including any parking allocated to the project that is outside the project boundary (see *LT Overview, Total Vehicle Parking Capacity*).

STEP 2. CALCULATE NUMBER OF PREFERRED PARKING SPACES AND ALTERNATIVE-FUEL FUELING STATIONS

Use Equation 1 to determine the number of preferred spaces required for green vehicles. Use Equation 2 for the number of fueling stations required. Projects must provide both preferred parking spaces and alternative-fuel fueling stations. In lieu of preferred parking, projects may provide discounted parking rates for all green vehicles (see *Further Explanation, Discounted Parking Rates*).

EQUATION 1. Required number of preferred parking spaces

$$\text{Preferred spaces} = \text{Total parking spaces} \times .05$$

EQUATION 2. Required number of fueling stations

$$\text{Fueling stations} = \text{Total parking spaces} \times .02$$

STEP 3. INCORPORATE PREFERRED PARKING INTO DESIGN

On the site plan, identify the location of preferred parking spaces for green vehicles (see *LT Overview, Preferred Parking*, and *Further Explanation, Examples*).

1. *epa.gov/climatechange/ghgemissions/sources/transportation.html* (*accessed June 10, 2013*).
2. *cta.ornl.gov/data/chapter11.shtml*, *Table 11.7* (*accessed June 10, 2013*).
3. *epa.gov/cleanschoolbus/antiidling.htm* (*accessed June 10, 2013*).
4. *cleanairactionplan.org/civica/filebank/blobdload.asp?BlobID=2516* (*accessed June 10, 2013*).

STEP 4. SELECT ALTERNATIVE-FUEL FUELING STATIONS

Consider legal, technical, and safety issues associated with each fuel type. Projects selecting electric vehicle supply equipment (EVSE) must follow Option 1. Projects selecting all other alternative fuel types must follow Option 2.

- Survey future building occupants to determine which alternative fuel is in highest demand. Regional availability of a given fuel type may be a factor.
- Compare the equipment associated with alternative fuels. The cost and complexity of designing and installing the fueling station vary with the type of fuel.
- Learn about the safety and maintenance issues associated with alternative fuels. Building personnel need to be trained to operate and maintain the fueling stations.
- Review local codes and standards for fueling facilities to determine whether other requirements must be met.

STEP 5. CONFIRM COMPLIANCE OF ANY ELECTRIC VEHICLE SUPPLY EQUIPMENT

For Option 1, select EVSE that meets the credit requirements for charge capability, standard compliance, and network features; all three must be documented for credit compliance. This information can often be found on the manufacturer's website.

- Only Level 2 and 3 chargers qualify for this credit.
- The SAE J1772 standard is common among U.S.-based EVSE manufacturers. IEC 62196 of the International Electrotechnical Commission is an equivalent standard that is common outside the U.S.
- The EVSE must have a Wi-Fi, ethernet, cellular modem, or other Internet communication mechanism that allows the device to send usage data to a server. This enables participation in demand response programs (e.g., SmartGrid) or time-of-use pricing. Specifications must demonstrate, at minimum, that the product is IP-addressable.
- A direct connection between the EVSE and on-site renewable energy is acceptable in lieu of network capability if the EVSE will draw solely from that energy source. Documentation must be provided.

STEP 6. INSTALL ALTERNATIVE-FUEL FUELING STATIONS

On the site plan, identify the location of each alternative-fuel fueling station (see *Further Explanation, Examples*).

- For Option 1, EVSE for plug-in electric vehicles or battery switching stations may be located in any area of the parking facility. Parking spaces reserved for EVSE may not contribute toward preferred parking spaces for green vehicles.
- For Option 2, liquid or gas fueling stations should be located outdoors if possible. If a liquid or gas fueling station cannot be located outdoors, provide dedicated exhaust connected directly to the outdoors.

STEP 7. ENSURE EFFECTIVE USE OF PARKING SPACES

Provide clear and permanently installed signage or pavement markings to reserve the preferred parking spaces for green vehicles only (see *Further Explanation, Signage for Green Vehicles*). In addition, clearly indicate the type of fueling station provided. For projects pursuing Option 1, provide signage that the parking space is reserved for plug-in electric vehicles only.

Projects providing discounted parking in lieu of preferred parking must communicate the discounted rate to users (see *Further Explanation, Discounted Parking Rates*).

Work with the property manager or parking management to review the policy for green vehicle parking and alternative-fuel facilities and ensure that the correct vehicles are using these spaces. Enforcement strategies will vary by project but should include consequences for violations by building users.

SCHOOLS

Determine which option the project will pursue.

Option 1 is best for schools whose vehicle fleets are not within reasonable control of the school management, and for those without vehicle fleets.

Option 2 is best for schools that are served by an outdated aged vehicle fleet and if the school management, project team, or owner has reasonable control over the vehicle fleet serving the building.

Option 1. Green Passenger Vehicles

Follow Steps 1 through 6 above; all references to Option 1 and Option 2 equate to Path 1 and Path 2 for Schools, respectively.

Option 2. Green Buses and School-Owned Vehicles

STEP 1. INVENTORY SCHOOL FLEET

Work with the school's fleet manager to identify the make, model, and year of each bus and each non-bus vehicle that serves the project building. All vehicles that serve the project building must be included in the inventory, including privately owned and leased vehicles.

- If the project building is part of a campus with a shared fleet, the inventory must include all buses in the campus bus fleet.
- If the project building has a dedicated bus fleet, inventory only the buses in the dedicated fleet.

STEP 2. EVALUATE BUS FLEET EMISSIONS

- Review manufacturers' data for each vehicle to find the NO_x and PM emissions values and identify each bus that exceeds the emissions in the credit requirements.
- If manufacturers' data are not available or do not include the emissions information, estimate NOx and PM emissions based on the vehicles' engine specifications and model years, using Table 1.
- Project teams outside the U.S. should use a local equivalent table of estimated emissions by model year, if one exists, or refer to manufacturers' data. Include the data source and any supporting calculations with the submittal documentation.

TABLE 1. Estimated emissions, by vehicle model year

Model Year	NO_x (g/bhp-hr)	PM (g/bhp-hr)
1974–1978	10.7	0.60
1979–1984	10.7	0.60
1985–1987	10.7	0.60
1988–1989	10.7	0.60
1990	6.0	0.60
1991–1993	5.0	0.25
1994–1997	5.0	0.10
1998–2003	4.0	0.10
2004–2006	4.0	0.10
2007 ON	0.2	0.01

Source: EPA's historical exhaust emissions standards for heavy-duty highway compression-ignition engines and urban buses: epa.gov/otaq/standards/heavy-duty/hdci-exhaust.htm.

STEP 3. EVALUATE VEHICLES OTHER THAN BUSES

Evaluate each vehicle to determine whether it qualifies as a green vehicle, per the rating system definition.

STEP 4. IDENTIFY PHASE-OUT AND RETROFIT STRATEGIES

If the fleet includes buses or other vehicles that do not meet credit requirements, determine whether to retrofit or phase out these vehicles to meet the emissions standards.

- Buses should be retrofitted using emissions control strategies approved by a relevant third party, such as the California Air Resources Board or a local equivalent. Confirm that each bus will meet credit emissions criteria by applying the emissions control strategies to the manufacturer's data or Table 1 estimates.

- Non-bus vehicles that do not meet green vehicle criteria must be phased out.

STEP 5. DOCUMENT PLANS TO ADDRESS ALL NONCOMPLIANT VEHICLES

Develop a plan to retrofit or phase out all noncompliant vehicles within seven years of the school's certificate of occupancy date. Include the following in the plan:

- Emissions evaluation of each vehicle in the current bus fleet serving the school
- Phase-out strategy or retrofit products for each bus that does not meet the emissions criteria
- Make, model and year of each non-bus vehicle, and whether it meets the green vehicle criteria
- Phase-out strategy for each non-bus vehicle that is not a green vehicle, including types of replacement vehicles
- Timeline and responsible parties for each stage of the plan
- For projects in which bus service is contracted, the policy governing scheduled contact bidding or unscheduled changes to the bus company contract
- Signed commitment from the school (or other entity with control over the fleet) to implement the plan

If bus or non-bus vehicle fleets used by the school already meet the credit requirements, supply the emissions evaluations for buses as well as make, model, and year information for non-bus vehicles.

WAREHOUSES AND DISTRIBUTION CENTERS

Option 1. Alternative-Fuel Vehicles

STEP 1. SELECT ALTERNATIVE-FUEL YARD TRACTOR

Purchase at least one yard tractor to be used at the project site facilities that is powered by electricity, propane, or natural gas.

- Before selecting a model, compare the environmental and economic costs and benefits of each fuel type. Regional availability of a given fuel type may also be a factor.
- Review local codes and standards for the type of fueling station that would be required.
- Compare the equipment associated with alternative fuels. The cost and complexity of designing and installing the fueling station vary with the type of fuel.
- Learn about the safety and maintenance issues associated with alternative fuels. Building personnel need to be trained to operate and maintain the fueling stations.
- If the project owner already owns an alternative-fuel yard tractor that will be dedicated to the facility, it is not necessary to purchase a new yard tractor to comply with the credit requirements.

STEP 2. PROVIDE ALTERNATIVE-FUEL FUELING STATION TO SUPPORT YARD TRACTOR(S)

- Work with the owner, civil engineer, mechanical engineer, and other team members to select an appropriate location for the alternative-fuel fueling station(s).
- Liquid or gas fueling stations should be located outdoors if possible. If a liquid or gas fueling station cannot be located outdoors, provide dedicated exhaust connected directly to the outdoors.

Option 2. Reduced Truck Idling

Install electrical connectors for at least 50% of loading dock doors. Indicate the locations on an electrical plan.

- When identifying the placement of electrical connectors, consider such factors as the frequency with which the door is used, the type of product carried at the door, the length of time trucks are idling, and project design elements that may require a reduction in engine idling (e.g., proximity to places of congregation or building entrances).
- Electrical connectors must allow drivers to shut down truck engines while maintaining use of in-cab heating, air-conditioning, communications, and entertainment systems.
- Although demonstrating this compatibility is not required, vehicles supplying the project will need adapters to take advantage of the electrical connectors. Contact rebate programs such as the Shorepower Truck Electrification Project in the U.S. (the-step-project.org) for potential assistance in subsidizing or supplying adapter kits.

FURTHER EXPLANATION

CALCULATIONS

See calculations in *Step-by-Step Guidance*.

DISCOUNTED PARKING RATES

Discounted parking rates for green vehicles may be used to achieve this credit in lieu of preferred parking if the following requirements are met:

- The discount must be at least 20%.
- The discount policy must be publicly posted at the entrance to the parking area and any other locations that may serve to inform building users of the discount. The discount must also be included in building policy documents and occupant advertisements.
- Projects cannot limit the discounted parking passes available; the discounted parking rate must be available to all building users who drive green vehicles.
- The discounted rate must be made available to all building users in perpetuity after the project's completion.

This approach is best applied when a large share of the parking capacity is devoted to monthly or yearly parking passes and car owners can receive the discounted rate when buying a pass. The application should ask for the car make and model so that the building operator can verify its compliance with the LEED definition of a green vehicle. Ensure that building personnel responsible for collecting parking fees are aware of the discount policy and the vehicles that are eligible.

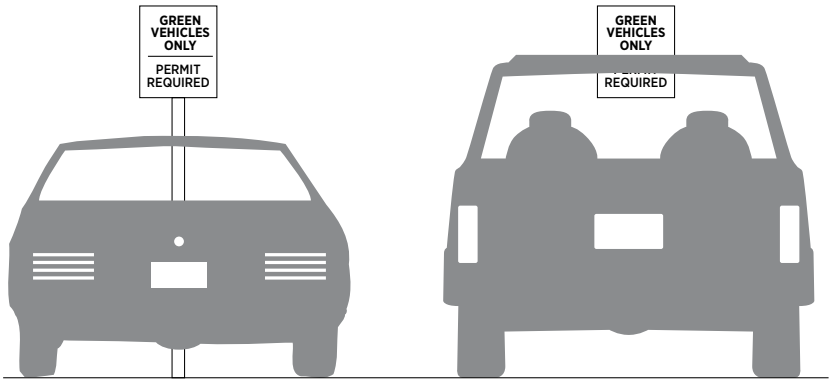
SIGNAGE FOR GREEN VEHICLES

Language used on reserved parking signage must reflect the spirit of the American Council for an Energy Efficient Economy (ACEEE) Green Book methodology (or a local equivalent for projects outside the U.S.).

In addition to this basic language, project teams are encouraged to share the definition of a green vehicle per the ACEEE Green Book (or a local equivalent for projects outside the U.S.). This information may be included with the preferred parking signage, near the preferred parking area, at the parking attendant booth or building entrance, or in educational material readily available to building users.

If a local green vehicle code or standard requires specific signage, this alternative language may be used, provided the standard has been approved by USGBC as equivalent to the ACEEE Green Book (see *International Tips*).

Figure 1. Example signage for green vehicle reserved parking



EXAMPLES

Example 1. Location of preferred parking and alternative-fuel fueling stations (single lot)
A new construction project building has a total parking capacity of 335 spaces. The project calculates 5% of the total to determine the number of preferred parking spaces:

335 total spaces × .05 = 16.75 preferred parking spaces

The result is rounded up to 17 preferred parking spaces. The project team decides to install EVSE in parking spaces reserved for plug-in electric vehicles and calculates the required spaces:

335 total spaces × .02 = 6.7 plug-in spaces

The team must provide seven plug-in spaces. Figure 2 shows the arrangement of preferred parking spaces and alternative-fuel fueling spaces.

Figure 2. Example of preferred parking locations



Example 2. Location of preferred parking in separated parking areas

A new construction project is a mixed-use building with a ground-level grocery store and office floors above. The parking area includes 108 spaces with separate parking areas for long- and short-term parking. Short-term parking, intended for the grocery store customers, numbers 36 spaces; long-term parking for office occupants numbers 72 spaces. The project team calculates the required preferred parking spaces as 5% of the total:

108 total spaces × .05 = 5.4 preferred parking spaces

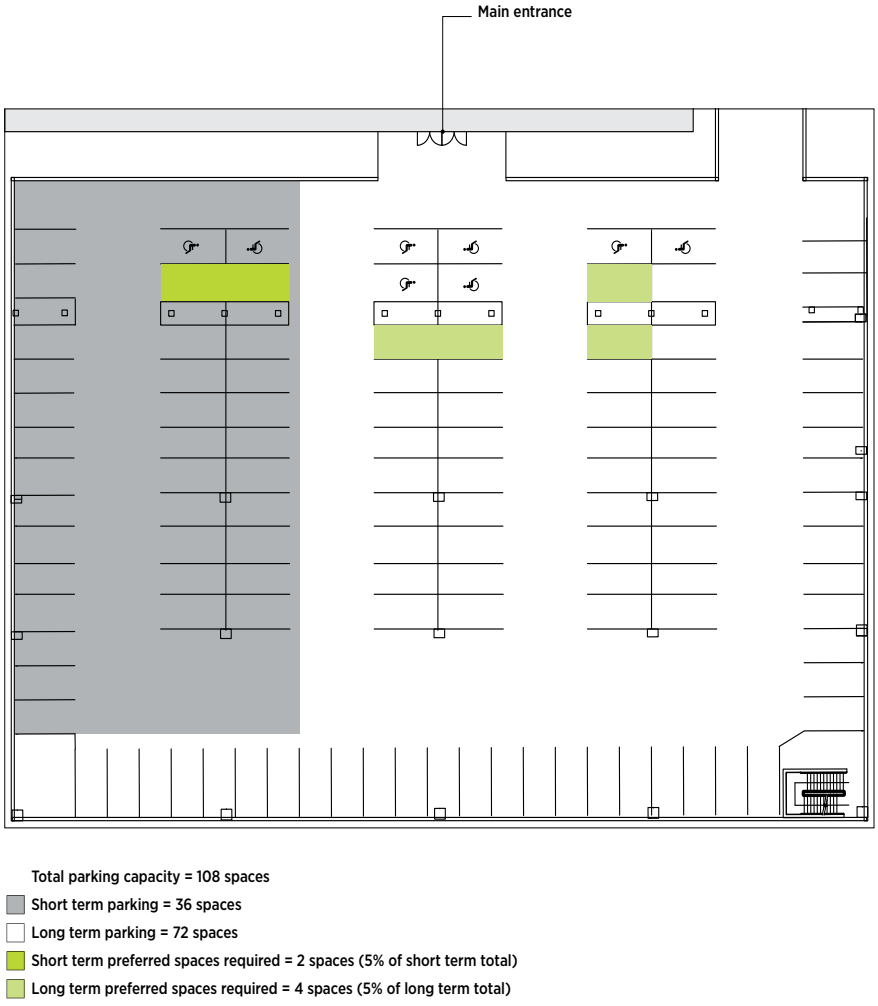
The result is rounded up to six preferred parking spaces. The project team must proportionally distribute these six preferred parking spaces between the short- and long-term parking areas. Because short-term parking accounts for one-third and long-term for two-thirds of the total parking, two short-term spaces (one-third of six) and four long-term spaces (two-thirds of six) are required. Figure 3 illustrates the result.

In addition, the project is providing natural gas fueling facilities. To determine the number of alternative-fuel vehicles to accommodate per day, the team makes the following calculation:

108 total spaces × .02 = 2.16 vehicles to be supported by refueling facilities

The result is rounded up. Thus the refueling facilities must have sufficient volume and rate to support three vehicles per day.

Figure 3. Example of proportionate distribution of preferred parking spaces



PROJECT TYPE VARIATIONS

Military Installations
Parking for ranking officials may be separated from regular parking, but the proportion of preferred parking for carpools must be applied to each pool of parking.

Separated Employee or Visitor Parking
Projects with separate parking areas for visitors, employees, or students must apply the proportion of preferred parking for carpools to each pool of parking.

INTERNATIONAL TIPS

Local equivalent standards to the American Council for an Energy Efficient Economy's (ACEEE) Green Book must comprehensively address vehicle fuel economy and vehicle emissions ratings, including particulate matter (PM), nitrogen oxides (NO_x), hydrocarbons, and carbon monoxide (CO). Complete and submit a side-by-side comparison of the selected local standard and ACEEE methodologies.

CAMPUS

Group Approach
New Construction, Core and Shell, Data Centers, Hospitality, Retail, Healthcare, Schools
All buildings in the group may be documented as one. All the parking located within the LEED project boundary must be included in the calculations. Submit a site plan that illustrates a reasonable distribution of preferred parking spaces for the buildings or spaces seeking LEED certification.

Warehouses and Distribution Centers
Option 1. All buildings in the group may be documented as one.
Option 2. Submit separate documentation for each building.

Campus Approach
New Construction, Core and Shell, Data Centers, Hospitality, Retail, Healthcare, Schools
Eligible. Submit a site plan that illustrates a reasonable distribution of preferred parking spaces for the projects seeking LEED certification.

Warehouses and Distribution Centers
Option 1. Eligible.
Option 2. Ineligible. Each LEED project may pursue the credit individually.

REQUIRED DOCUMENTATION

Documentation	New Construction, Core and Shell, Data Centers, Hospitality, Retail, Healthcare		Schools	
	Option 1	Option 2	Option 1	Option 2
Parking or site plan indicating main building entrance, preferred parking spaces, and alternative-fuel fueling stations; calculations based on total parking capacity	X	X	X	
For preferred parking spaces, photographs of signage or pavement marking	X	X	X	
For electric vehicle charging spaces, photographs of signage or pavement marking	X		X	
For discounted parking rate, copy of communication to building occupants or photograph of signage	X	X	X	
For electrical connectors, manufacturers' product specifications indicating charge level, compliance with relevant standard, and Internet addressability	X		X	
For liquid or gas fueling stations, manufacturers' product specifications indicating fuel type and refueling rate		X	X	
Phase-in plan for emissions-compliant bus fleet, including emissions evaluation of current fleet, retrofit strategies, timeline, responsible parties				X
Phase-in plan for green non-bus vehicles, including types of vehicles, timeline and responsible parties				X

Warehouses and Distribution Centers		
Documentation	**Option 1**	**Option 2**
Manufacturer's documentation of yard tractor model and fuel type	X	
Site plan showing electrical connector locations at loading dock doors		X
Manufacturer's documentation for electrical connectors indicating ability to power in-cab amenities		X

RELATED CREDIT TIPS

LT Credit Reduced Parking Footprint. Project teams also pursuing the related credit must reserve 12% of total parking capacity for preferred vehicles (with a required 7% for green vehicles in LT Credit Green Vehicles and 5% for carpools or vanpools in LT Credit Reduced Parking Footprint). Guidance for locating preferred parking is the same for both credits.

EA Credit Demand Response. Any electric charging stations should align with the existing demand response program or infrastructure to comply with the related credit.

EQ Prerequisite Minimum Air Quality Performance. Consider separate ventilation for liquid or gas alternative-fuel fueling facilities alongside development of the mechanical plan and indoor air quality strategy to comply with the related prerequisite.

CHANGES FROM LEED 2009

New Construction, Core and Shell, Data Centers, Hospitality, Retail, Healthcare

- The term "low-emitting and fuel-efficient vehicles" has been changed to "green vehicles."
- Options 1 and 2 have been reorganized such that all projects must provide preferred parking for green vehicles and alternative-fuel fueling stations.
- The minimum ACEEE green score, which determines classification as a fuel-efficient vehicle, has been raised to 45.
- The requirement for alternative-fuel stations has been lowered to 2% of total parking capacity.
- Electrical connectors must comply with SAE Surface Vehicle Recommended Practice J1772, SAE Electric Vehicle Conductive Charge Coupler (or a regional or local equivalent standard) and must also be capable of dynamic interaction with the utility grid.
- Credit can no longer be earned by providing green vehicles or a car-sharing program for building occupants.

Schools

- See section above for changes in Option 1.
- A designated carpool drop-off area for green vehicles (formerly "low-emitting and fuel-efficient vehicles") is no longer required.
- Option 2 now requires an implementation plan to meet NO_x and particulate emissions standards for school buses and green vehicle designation for vehicles other than buses.

Warehouses and Distribution Centers

- Credit requirements specific to warehouses and distribution centers have been added.

REFERENCED STANDARDS

American Council for an Energy Efficient Economy (ACEEE) Green Book: greenercars.org

Society of Automotive Engineers, SAE Surface Vehicle Recommended Practice J1772, SAE Electric Vehicle Conductive Charge Coupler: standards.sae.org/j1772_201001

International Electrical Commission 62196: iec.ch

EXEMPLARY PERFORMANCE

Not available.

DEFINITIONS

alternative fuel low-polluting, nongasoline fuels such as electricity, hydrogen, propane, compressed natural gas, liquid natural gas, methanol, and ethanol

demand response (DR) a change in electricity use by demand-side resources from their normal consumption patterns in response to changes in the price of electricity or to incentive payments designed to induce lower electricity use at times of high wholesale market prices or when system reliability is jeopardized

electric vehicle supply equipment the conductors, including the ungrounded, grounded, and equipment grounding conductors, the electric vehicle connectors, attachment plugs, and all other fittings, devices, power outlets or apparatuses installed specifically for the purpose of delivering energy from the premises wiring to the electric vehicle. (National Electric Codes and California Article 625)

grams per brake horsepower hour metric used to communicate how many grams of emissions (e.g., nitrogen oxide or particulate matter) are emitted by an engine of a specific horsepower rating over a one-hour period

preferred parking the parking spots closest to the main entrance of a building (exclusive of spaces designated for handicapped persons). For employee parking, it refers to the spots that are closest to the entrance used by employees.

time-of-use pricing an arrangement in which customers pay higher fees to use utilities during peak time periods and lower fees during off-peak time periods

yard tractor a vehicle used primarily to facilitate the movement of truck trailers and other types of large shipping containers from one area of a site to another. It does not include forklift trucks. Also known as terminal tractor, yard truck, utility tractor rig, yard goat, or yard hustler.